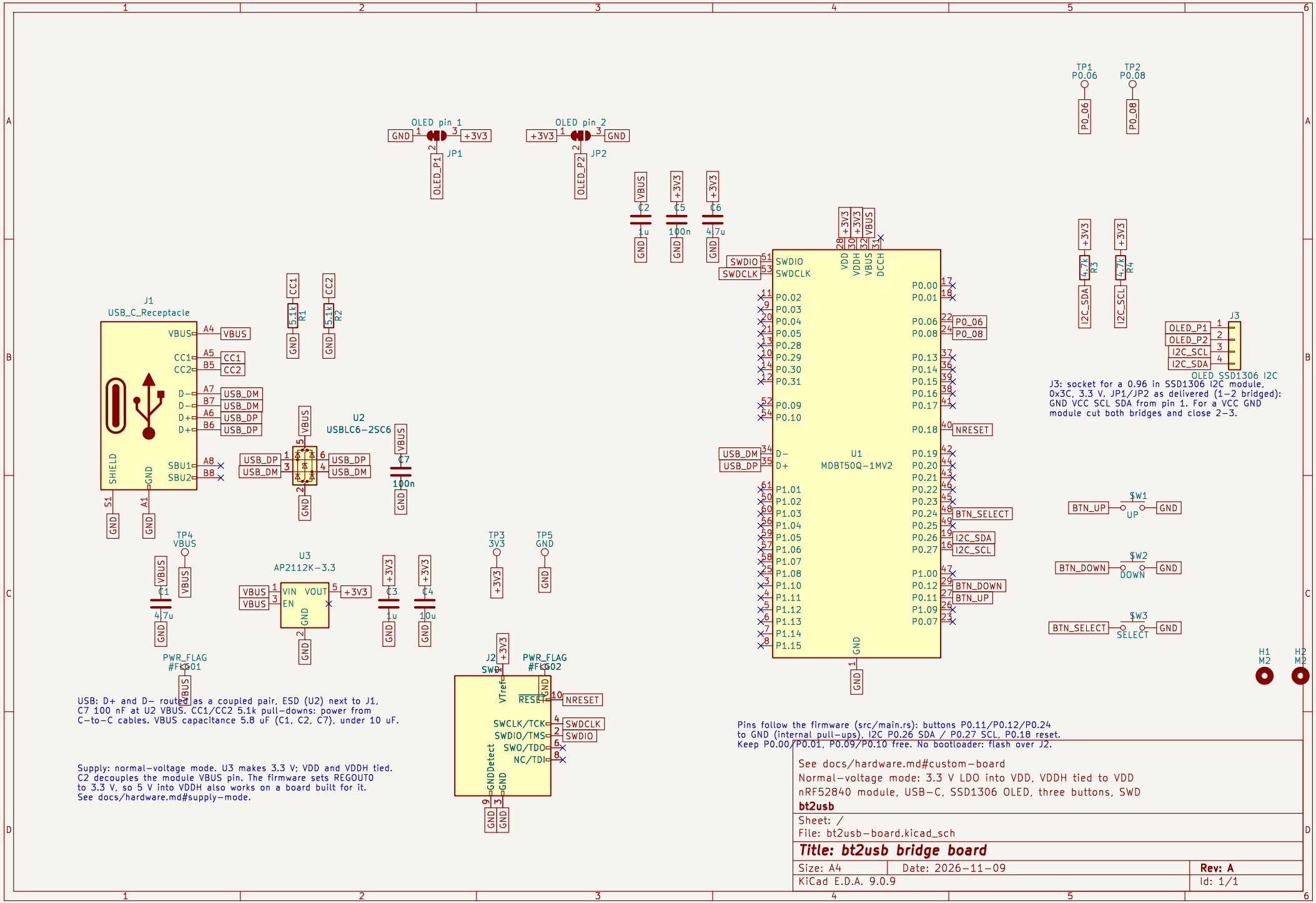

J1
USB_C_Receptacle
VBUS
CC1
CC2
D−
D−
D+
D+
SBU1
SBU2
SHIELD
GND
S1
A1
A4
A5
B5
A7
B7
A6
B6
A8
B8
USB_DM
USB_DM
USB_DP
USB_DP
CC1
CC2
5.1k
R1
5.1k
R2
U2
USBLC6−2SC6
C7
100n
U3
AP2112K−3.3
VIN
VOUT
EN
GND
C1
4.7u
C3
1u
C4
10u
TP4
VBUS
TP3
3V3
TP5
GND
PWR_FLAG
#FLG01
PWR_FLAG
#FLG02
OLED pin 1
JP1
OLED pin 2
JP2
OLED_P1
OLED_P2
C2
1u
C5
100n
C6
4.7u
U1
MDBT50Q−1MV2
SWDIO
SWDCLK
VDD
VDDH
VBUS
DCCH
P0.00
P0.01
P0.02
P0.03
P0.04
P0.05
P0.28
P0.29
P0.30
P0.31
P0.09
P0.10
P0.06
P0.08
P0.13
P0.14
P0.15
P0.16
P0.17
P0.18
NRESET
P0.19
P0.20
P0.21
P0.22
P0.23
P0.24
P0.25
P0.26
P0.27
P1.01
P1.02
P1.03
P1.04
P1.05
P1.06
P1.07
P1.08
P1.10
P1.11
P1.12
P1.13
P1.14
P1.15
P1.00
P0.12
P0.11
P1.09
P0.07
BTN_SELECT
I2C_SDA
I2C_SCL
BTN_DOWN
BTN_UP
TP1
P0.06
TP2
P0.08
4.7k
R3
4.7k
R4
J3
OLED_P1
OLED_P2
I2C_SCL
I2C_SDA
OLED SSD1306 I2C
J3: socket for a 0.96 in SSD1306 I2C module,
0x3C, 3.3 V. JP1/JP2 as delivered (1−2 bridged):
GND VCC SCL SDA from pin 1. For a VCC GND
module cut both bridges and close 2−3.
BTN_UP
SW1
UP
GND
BTN_DOWN
SW2
DOWN
GND
BTN_SELECT
SW3
SELECT
GND
H1
M2
H2
M2
J2
SWD
VTref
GNDDetect
RESET
SWCLK/TCK
SWDIO/TMS
SWO/TDO
NC/TDI
NRESET
SWDCLK
SWDIO
USB: D+ and D− routed as a coupled pair, ESD (U2) next to J1,
C7 100 nF at U2 VBUS. CC1/CC2 5.1k pull−downs: power from
C−to−C cables. VBUS capacitance 5.8 uF (C1, C2, C7), under 10 uF.
Supply: normal−voltage mode. U3 makes 3.3 V; VDD and VDDH tied.
C2 decouples the module VBUS pin. The firmware sets REGOUT0
to 3.3 V, so 5 V into VDDH also works on a board built for it.
See docs/hardware.md#supply−mode.
Pins follow the firmware (src/main.rs): buttons P0.11/P0.12/P0.24
to GND (internal pull−ups), I2C P0.26 SDA / P0.27 SCL, P0.18 reset.
Keep P0.00/P0.01, P0.09/P0.10 free. No bootloader: flash over J2.
See docs/hardware.md#custom−board
Normal−voltage mode: 3.3 V LDO into VDD, VDDH tied to VDD
nRF52840 module, USB−C, SSD1306 OLED, three buttons, SWD
bt2usb
Sheet: /
File: bt2usb−board.kicad_sch
Title: bt2usb bridge board
Size: A4
Date: 2026−11−09
Rev: A
KiCad E.D.A. 9.0.9
Id: 1/1